

# Regularizers, LMI constraints, and constraint (non)degeneracy

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This note expands on which regularizers / LMI constraints can be added to the inner SDP while keeping the envelope-theorem assumptions **A1–A4**, and then explains in detail **what constraint (non)degeneracy is** and **why degeneracy breaks the exact gradient**.

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## 1. Recap — the admissible class

Split additions by *where* they live.

- **Design-cost side**  $J_{\text{des}}(\theta)$ : any convex penalty on  $\theta$  (sparsity, group-sparsity, budgets). These never touch the inner SDP's A1–A4; nonsmooth ones are handled by the outer prox/epigraph. Always compatible.
- **Inner SDP side**: A1 (convex,  $C^1$  data after conic lifting, Slater, attainment) and A4 ( $\theta$  enters through finitely many  $C^1$  data) admit essentially **any convex, conic-representable term whose data are  $C^1$  in  $\theta$  and that keeps a strictly feasible point**: trace terms, gain norms, nuclear norm, log-det/Tikhonov, variance/output/ellipsoid caps, D-stability (pole-region) LMIs, polytopic quadratic stability, gain-scheduling/LPV LMIs, passivity/IQC, structural gain patterns.

**A2 and A3 are the gate.** They are *not* automatic under convex additions. Two universal recipes make them hold for the whole family at once:

1. **A strictly convex regularizer** in the variables — a small Tikhonov  $\varepsilon \|\cdot\|^2$  or  $-\varepsilon \log \det \Sigma$  — makes the inner objective strictly convex, forcing **A2** (unique primal) and generically restoring A3.
2. **Nondegenerate active constraints** — the subject of the rest of this note — give **A3** (unique dual, differentiable value).

The augmented envelope gradient simply gains one term per constraint,  $\nabla_{\theta} V = \partial_{\theta} \phi + \sum_i \langle \lambda_i^*, \partial_{\theta}(\text{data}_i) \rangle$ .

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## 2. What the assumptions really need

Restated in the language that matters for perturbation:

- **A2** = the primal minimizer is unique.
- **A3** = the dual multiplier is unique **and** strict complementarity holds, so the optimal value  $V(\theta)$  is differentiable and  $\nabla V = \partial_{\theta} \mathcal{L}$  at that single multiplier.

The classical SDP sensitivity results (Alizadeh–Haeberly–Overton 1997; Shapiro 1997; Bonnans–Shapiro 2000) tie these to three geometric properties of the primal–dual pair:

**dual nondegeneracy**  $\Rightarrow$  unique primal (A2),      **primal nondegeneracy**  $\Rightarrow$  unique dual (A3),  
**strict complementarity** + nondegeneracy  $\Rightarrow V$  differentiable.

So "constraint nondegeneracy" is exactly the property that buys A3 (and its dual buys A2).

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### 3. What "nondegenerate" means

#### 3a. Scalar inequality constraints — LICQ

For a program  $\min_z f(z)$  s.t.  $g_i(z) \leq 0$ , let  $\mathcal{A} = \{i : g_i(z^*) = 0\}$  be the **active set** at the optimum. The **linear-independence constraint qualification (LICQ)** is:

$$\{\nabla g_i(z^*)\}_{i \in \mathcal{A}} \text{ are linearly independent.}$$

Under LICQ the KKT multiplier  $\lambda^*$  is **unique**. LICQ *failing* — active constraint gradients linearly dependent — is exactly (scalar) degeneracy.

#### 3b. LMI / conic constraints — primal & dual nondegeneracy

For an LMI  $M(z) \succeq 0$  that is active at  $z^*$  (i.e.  $M(z^*)$  is singular), degeneracy is more subtle because the "active set" is a *face* of the PSD cone, not a finite index set. Let the columns of  $Q$  span  $\ker M(z^*)$  (dimension = the rank drop). Following Alizadeh–Haeberly–Overton / Shapiro:

- **Primal nondegeneracy:** the linear map  $z \mapsto Q^\top (D_z M(z^*)) Q$  is **surjective** onto the symmetric matrices on  $\ker M$ . Intuitively: the constraint's directions "reach" every way the kernel could move — the active face is hit *transversally*. This is the LMI analogue of LICQ, and it implies the **dual is unique**.
- **Dual nondegeneracy:** the analogous surjectivity on the dual side; it implies the **primal is unique**.
- **Strict complementarity:** the primal slack  $M(z^*)$  and its multiplier  $S^*$  have complementary full rank,

$$\text{rank } M(z^*) + \text{rank } S^* = \text{dim}.$$

Their ranges are orthogonal complements (no "shared" kernel direction).

A constraint is **degenerate** if any of these fails: dependent active directions (primal), or a rank overlap  $\text{rank } M^* + \text{rank } S^* < \text{dim}$  (loss of strict complementarity).

In the LQG appendix this is exactly why A3 goes through: at the optimum  $\text{rank } M_2^* = n$ , its multiplier has  $\text{rank } S^* = n$  (core  $\Xi = P \succ 0$ ), and  $n + n = 2n = \text{full}$  — strict complementarity holds, and the lifted form makes the dual a single point.

### 4. What a degenerate constraint looks like

Concrete, minimal examples.

**(i) Redundant / duplicated constraints (dependent gradients).** Add the same variance cap twice, or include a polytope vertex dominated by the others in a quadratic-stability set. At the optimum both are "active" but their normals are linearly dependent, so the multiplier splits arbitrarily — the dual is a line, not a point.

**(ii) A "tie" between two active constraints.** The canonical parametric example:

$$V(\theta) = \min_x x \quad \text{s.t.} \quad x \geq \theta_1, \quad x \geq \theta_2, \quad x^* = \max(\theta_1, \theta_2).$$

When  $\theta_1 = \theta_2$  both are active; the multiplier set is the whole simplex  $\{\lambda \geq 0 : \lambda_1 + \lambda_2 = 1\}$  — **non-unique** — and  $V(\theta) = \max(\theta_1, \theta_2)$  has a **kink** there. Away from the tie exactly one constraint is active, nondegenerate,  $V$  smooth. Degeneracy = the tie.

(iii) **SDP rank overlap (loss of strict complementarity).** The minimal- $\gamma$   $H_\infty$  synthesis LMI at  $\gamma = \gamma^*$ : the certificate makes the bounded-real matrix drop rank *and* the dual drops rank so that  $\text{rank } M^* + \text{rank } S^* < \text{dim}$ . The optimal face is not a vertex; the dual is non-unique and  $\gamma^*(\theta)$  is nonsmooth. This is example (ii) lifted to matrices (a "tie" between worst-case frequencies).

(iv) **Structural gain equalities.** Coupled decentralized- $K$  patterns can make the active equality constraints linearly dependent (a fixed-pattern gain that is already implied by others), again splitting the multiplier.

## 5. Why degeneracy is a problem

Everything the envelope theorem promises rests on a **unique** multiplier and a **smooth** value. Degeneracy destroys both:

1. **The dual is not a singleton.** With multiplier set  $\mathcal{D}^*(\theta)$  not a point, "read the multiplier and contract it against  $\partial_\theta(\text{data})$ " is **ambiguous** — different solvers (or the same solver on a different run) return different  $\lambda^* \in \mathcal{D}^*$ , giving different gradients.
2. **The value is only directionally differentiable.** In place of a gradient one has

$$V'(\theta; d) = \min_{\lambda \in \mathcal{D}^*(\theta)} \langle \partial_\theta \mathcal{L}(z^*, \lambda, \theta), d \rangle,$$

which is generally **not linear in  $d$**  (it's a min over a set), so  $V$  has **kinks** — exactly the  $\max(\theta_1, \theta_2) / H_\infty$ -norm nonsmoothness.  $\partial_\theta \mathcal{L}$  at any returned  $\lambda$  is then a **subgradient**, not the gradient.

3. **The solution map can be ill-posed.** Degeneracy usually coincides with a non-unique or non-Lipschitz primal (A2 also fails), so the sensitivity  $\partial z^* / \partial \theta$  need not even exist — the outer optimizer's secant information becomes unreliable and BFGS curvature degrades.

In short: **degeneracy turns the exact-gradient guarantee into a subgradient at best, and can make the very quantity you differentiate nonsmooth.** That is why A3 is the delicate assumption and why nondegeneracy is required to keep the framework exact.

## 6. Subgradients: what they are, when you get one, and why they help

What a subgradient is

For a convex function  $V : \mathbb{R}^p \rightarrow \mathbb{R}$ , a vector  $g$  is a **subgradient** at  $\theta$  if it defines a globally under-estimating affine minorant:

$$V(\theta') \geq V(\theta) + \langle g, \theta' - \theta \rangle \quad \forall \theta'.$$

The set of all such  $g$  is the **subdifferential**  $\partial V(\theta)$ . Geometrically, each  $g$  is the slope of a hyperplane that touches the graph of  $V$  at  $\theta$  and lies below it everywhere. Where  $V$  is differentiable,  $\partial V(\theta) = \{\nabla V(\theta)\}$  — a single element, the ordinary gradient. Where  $V$  has a **kink**,  $\partial V(\theta)$  is a whole set (an interval/polytope of admissible slopes).

The connection to the directional derivatives of §5 is exact: the subdifferential is the set whose support function is the directional derivative,

$$V'(\theta; d) = \max_{g \in \partial V(\theta)} \langle g, d \rangle \quad (\text{convex case}),$$

so " $V$  has a kink" and " $\partial V$  is set-valued" are the same statement. (For a nonconvex but locally Lipschitz  $V$  — the general co-design value — the analogue is the **Clarke subdifferential**, the convex hull of limits of nearby gradients; the same picture holds locally.)

When you get a subgradient instead of a gradient

Precisely when  $V$  is **not differentiable** at  $\theta$  — i.e. exactly the degeneracy of §3–§5:

- **non-unique dual** (primal degeneracy / LICQ failure): the envelope map returns  $\partial_\theta \mathcal{L}(z^*, \lambda, \theta)$  for *some*  $\lambda \in \mathcal{D}^*$ , and each choice is a subgradient —

$$\partial V(\theta) = \{ \partial_\theta \mathcal{L}(z^*, \lambda, \theta) : \lambda \in \mathcal{D}^*(\theta) \};$$

- **loss of strict complementarity** (rank overlap): a kink along the offending direction;
- **active-set ties** (example (ii)), the **minimal- $\gamma$**   $H_\infty$  optimum (example (iii)), and nonsmooth regularizers used *without* the epigraph lift.

When A1–A3 hold (nondegenerate, strictly complementary),  $\mathcal{D}^*$  is a singleton, so  $\partial V(\theta) = \{\nabla V(\theta)\}$ : **the subgradient is the gradient**. Thus "subgradient vs. gradient" is exactly "degenerate vs. nondegenerate."

Why a subgradient is still useful

- **Optimization still works.** Projected/proximal **subgradient** and **bundle** methods converge on nonsmooth convex problems using *any*  $g \in \partial V$ ; the outer quasi-Newton loop degrades gracefully to such a step. You lose the superlinear rate, not correctness.
- **Optimality certificate.**  $\theta^*$  is stationary iff  $0 \in \partial V(\theta^*)$  (plus the box's normal cone) — a clean test that remains valid at kinks, where  $\nabla V$  does not exist.
- **A guaranteed descent direction exists.** The steepest-descent direction is  $-g^*$  with  $g^* = \arg \min_{g \in \partial V(\theta)} \|g\|$  (the minimum-norm subgradient); it points downhill whenever  $0 \notin \partial V$ .
- **It is what nonsmooth  $H_\infty$  synthesis already uses.** The minimal- $\gamma$  value is nonsmooth, and subgradient/bundle methods (HIFOO, Apkarian–Noll) are the standard, effective tools there — so the fallback is a well-trodden path, not a dead end.

So a subgradient is a principled continuation with a weaker (linear rather than superlinear) rate; it keeps the method usable even when a design sits on a degenerate face.

## 7. How to detect and ensure nondegeneracy

- **Rank test.** Compute the primal slack  $M(z^*)$  and its multiplier  $S^*$ ; strict complementarity  $\Leftrightarrow \text{rank } M^* + \text{rank } S^* = \text{dim}$ . A rank overlap flags degeneracy.
- **Remove redundancy.** Drop dominated polytope vertices, duplicate caps, and linearly dependent structural equalities before solving — these are the commonest sources of dependent active constraints.
- **Add a strictly convex regularizer** (recipe 1: Tikhonov / log-det). It pins the primal (A2) and, generically, perturbs the optimum off degenerate faces, restoring strict complementarity (A3). This is the cheapest universal safeguard.
- **Keep caps strictly feasible.** A variance/output cap set *above* the achievable floor is inactive or nondegenerately active; a cap pushed *to* the floor is the matrix analogue of a tie (example (ii)/(iii)) and should be avoided.
- **The finite-difference gradient check** is a practical degeneracy detector: if the analytic envelope gradient matches central differences (as in the studies, rel. err  $\sim 10^{-6}$ ), the point is nondegenerate; a

mismatch localizes a degenerate design.

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## 8. The graceful fallback

Where degeneracy is unavoidable at isolated designs, the method does **not** fail: the returned multiplier gives a valid **subgradient** (§6), and the projected/quasi-Newton outer loop degrades to a subgradient step (Remark on the degenerate case). Recipe 1 (a strictly convex regularizer) removes almost all such points, so in practice the exact-gradient regime holds everywhere the studies operate.

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## References

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